

Tal Wagner

CONTACT INFORMATION	talw@mit.edu http://www.mit.edu/~talw
RESEARCH INTERESTS	Algorithms for large and high-dimensional data, finite metric spaces, large-scale machine learning.
EDUCATION	MIT – Massachusetts Institute of Technology Ph.D. candidate in Computer Science - Advisor: Prof. Piotr Indyk Cambridge, MA 2014–2020 Weizmann Institute of Science M.Sc. in Computer Science and Mathematics - Advisor: Prof. Uriel Feige Rehovot, Israel 2011–2013 Technion - Israel Institute of Technology B.Sc. in Computer Science, <i>summa cum laude</i> B.Sc. in Mathematics, <i>summa cum laude</i> - Technion Excellence Program Haifa, Israel 2008–2011
PUBLICATIONS	Eccentricity Heuristics through Sublinear Analysis Lenses. 1st Symposium on Algorithmic Principles of Computer Systems (APoCS 2020). Space and Time Efficient Kernel Density Estimation in High Dimensions (with A. Backurs and P. Indyk). 33rd Annual Conference on Neural Information and Processing Systems (NeurIPS 2019). Scalable Fair Clustering (with A. Backurs, P. Indyk, K. Onak, B. Schieber and A. Vakilian). 36th International Conference on Machine Learning (ICML 2019). Sample-Optimal Low-Rank Approximation of Distance Matrices (with P. Indyk, A. Vakilian and D.P. Woodruff). 32nd Annual Conference on Learning Theory (COLT 2019). Semi-Supervised Learning on Data Streams via Temporal Label Propagation (with S. Guha, S. Kasiviswanathan and N. Mishra). 35th International Conference on Machine Learning (ICML 2018). Approximate Nearest Neighbors in Limited Space (with P. Indyk). 31st Annual Conference on Learning Theory (COLT 2018). Practical Data-Dependent Metric Compression with Provable Guarantees (with P. Indyk and I. Razenshteyn). 31st Annual Conference on Neural Information and Processing Systems (NeurIPS 2017). A Graph-Theoretic Approach to Multitasking (with N. Alon, D. Reichman, I. Shinkar, S. Musslick, B. Dey, K. Ozcimder, J.D. Cohen and T.L. Griffiths). 31st Annual Conference on Neural Information and Processing Systems (NeurIPS 2017). Near-Optimal (Euclidean) Metric Compression (with P. Indyk). 28th ACM-SIAM Symposium on Discrete Algorithms (SODA 2017).

10. **Cheeger-Type Approximation for Sparsest st-Cut** (with R. Krauthgamer).
ACM Transactions on Algorithms (TALG 2016).
11. **Towards Resistance Sparsifiers** (with M. Dinitz and R. Krauthgamer).
19th International Workshop on Randomization and Computation (RANDOM 2015).
12. **Volume Regularization for Binary Classification** (with K. Crammer).
26th Annual Conference on Neural Information and Processing Systems (NeurIPS 2012).

PREPRINTS

(AVAILABLE ONLINE)

1. **Scalable Nearest Neighbor Search for Optimal Transport** (with Y. Dong, P. Indyk and I. Razenshteyn).
Preliminary version accepted to the 3rd Optimal Transport in Machine Learning workshop (OTML) at NeurIPS 2019.
2. **Learning Space Partitions for Nearest Neighbor Search** (with Y. Dong, P. Indyk and I. Razenshteyn).
3. **Multitasking Capacity: Hardness Results and Improved Constructions** (with N. Alon, J.D. Cohen, T.L. Griffiths, P. Manurangsi, D. Reichman, I. Shinkar and A. Yu).

EMPLOYMENT

Microsoft Research Research intern in Machine Learning and Optimization group Host: Ilya Razenshteyn	Redmond, WA Summer 2019
Amazon Applied Scientist intern in Core Machine Learning group Host: Nina Mishra	Palo Alto, CA Summer 2017
VMware Research intern in VMware Research Group Host: Udi Wieder	Palo Alto, CA Summer 2015
Google Software engineer	Tel-Aviv, Israel 2013-2014

TEACHING

MIT – Massachusetts Institute of Technology <i>Advanced Algorithms</i> , teaching assistant.	Fall 2016
Technion - Israel Institute of Technology <i>Intro. to Computer Science</i> , teaching assistant.	2009-2010